

MA 403 Real Analysis

Homework 2

1. Find the sup and inf of the set S , where $S = \{x : 3x^2 - 10x + 3 < 0\}$.

2. Prove Lagrange's identity for real numbers:

$$\left(\sum_{k=1}^n a_k b_k\right)^2 = \left(\sum_{k=1}^n a_k^2\right) \left(\sum_{k=1}^n b_k^2\right) - \sum_{1 \leq k < j \leq n} (a_k b_j - a_j b_k)^2$$

Use this identity to deduce the Cauchy-Schwarz inequality.

3. Let $f : S \rightarrow T$ be a function. Prove that the following statements are equivalent:

(a) f is one-to-one on S .

(b) $f^{-1}(f(A)) = A$ for every subset A of S .

(c) For all subsets A and B of S with $B \subseteq A$, we have $f(A - B) = f(A) - f(B)$

4. Let S be the relation given by defining S to be the set of all pairs of real numbers (x, y) that satisfy the given equation (or inequality). Determine in each case, whether S is *reflexive*, or *symmetric*, or *transitive* (it may satisfy more than one of these conditions).

(a) $x \leq y$

(b) $x^2 + y^2 = 1$

5. Show that the following sets are countable:

(a) the set of circles in $\mathbb{R}^2$ having rational radii and centers with rational coordinates.

(b) any collection of disjoint intervals of positive length.

6. Is the set of all irrational real numbers countable? Explain your answer.